

ME5412 Robotics for Healthcare

SCI and assessment scales

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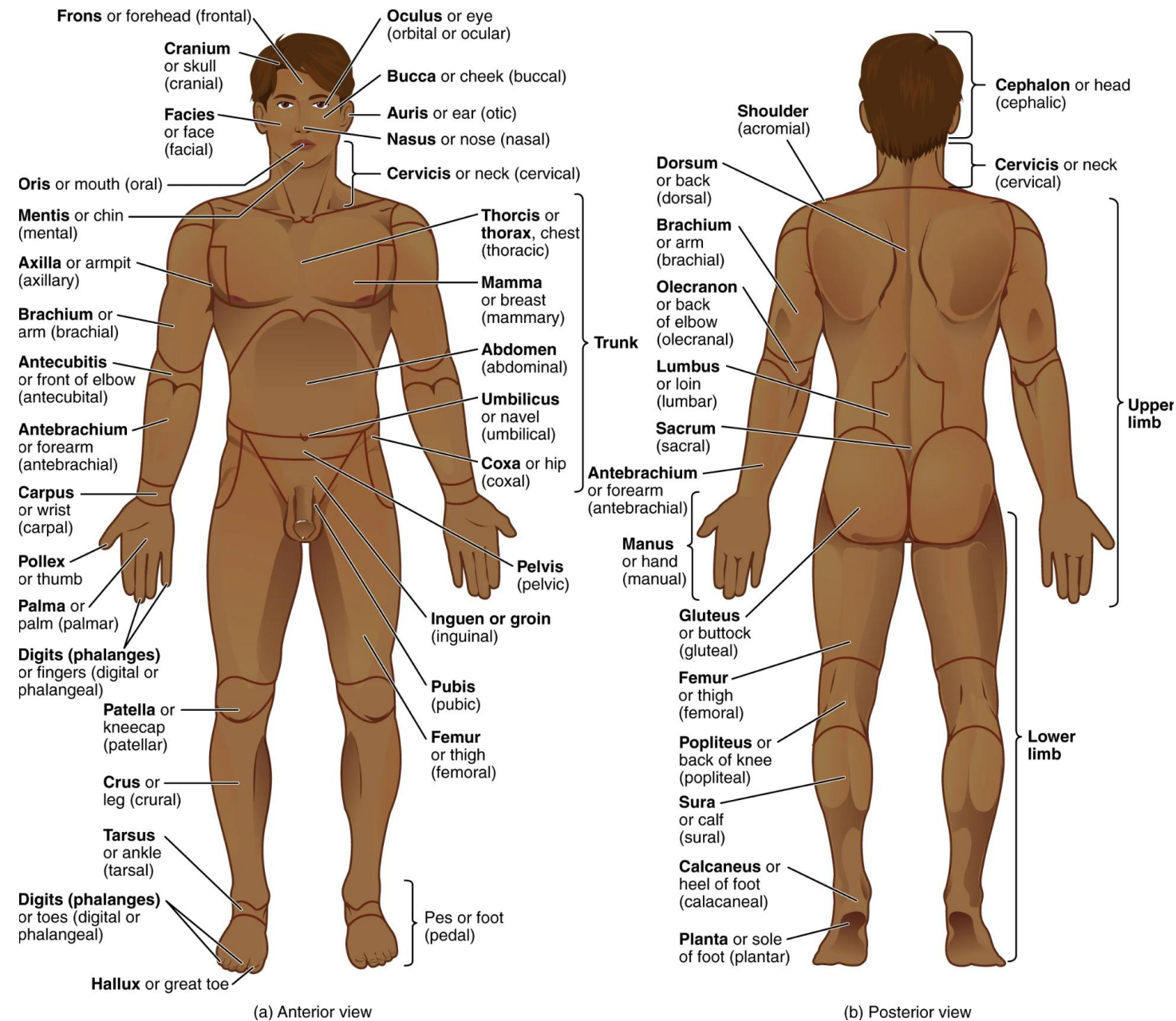
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Lecture outline

- Basic concepts and terms of human anatomy
- Spinal cord injuries
- Assessment scales

Regions of the human body



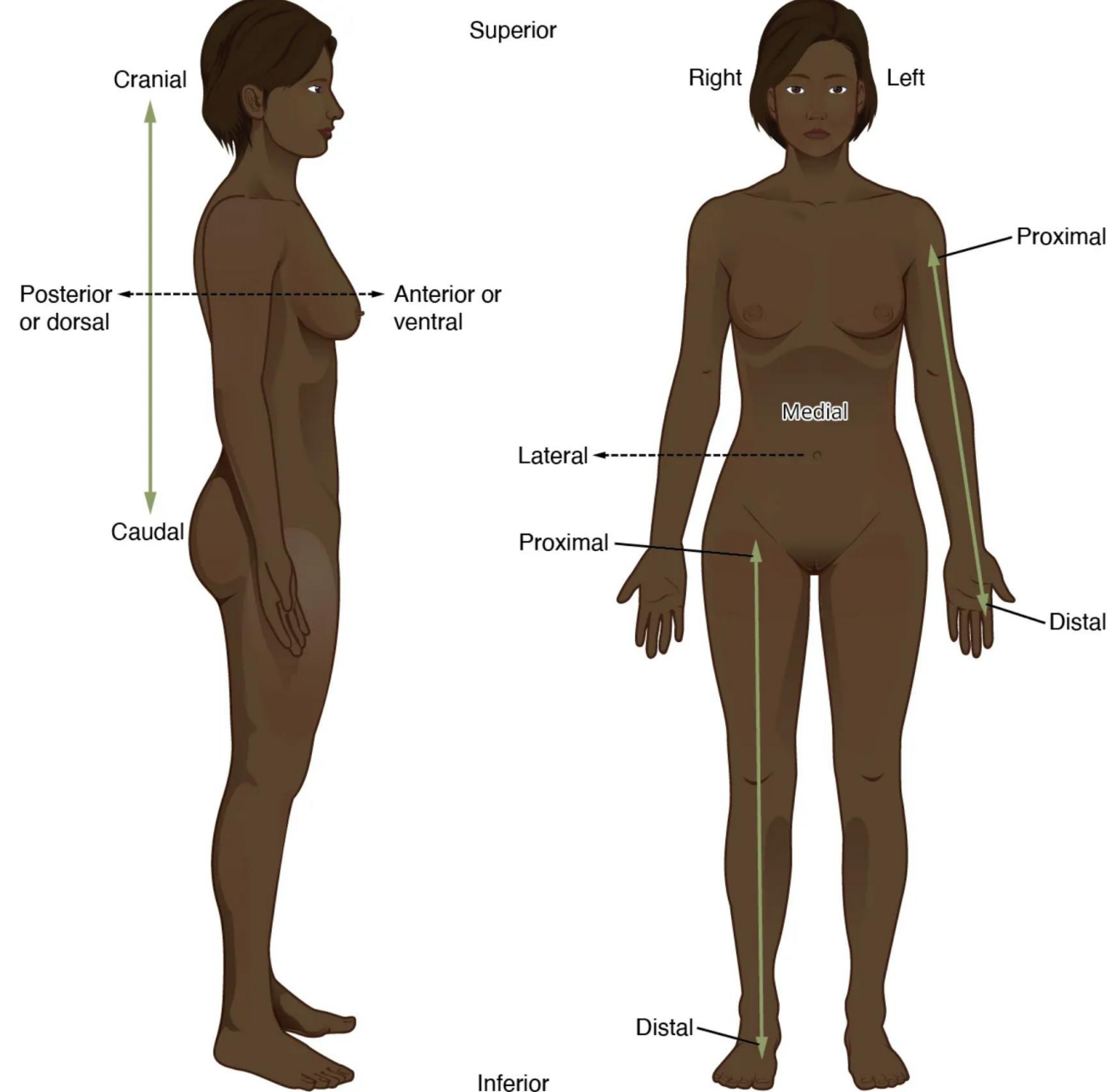
(a) Anterior view

(b) Posterior view

Directional Terms

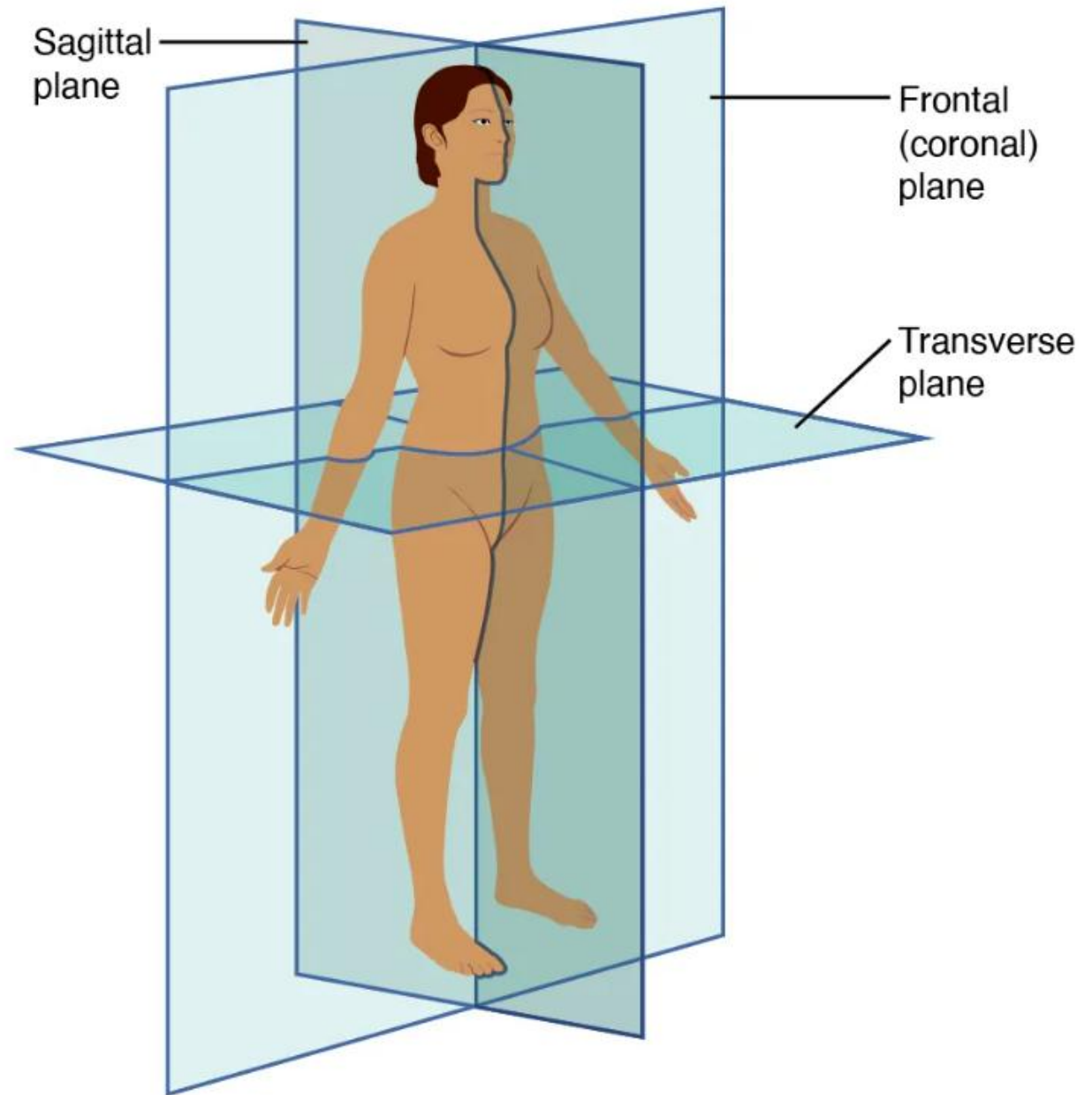
Terms for describing the relative locations of different body structures

- **Anterior** (or **ventral**) Describes the front or direction toward the front of the body. The toes are anterior to the foot.
- **Posterior** (or **dorsal**) Describes the back or direction toward the back of the body. The popliteus is posterior to the patella.
- **Superior** (or **cranial**) describes a position above or higher than another part of the body proper. The orbits are superior to the oris.
- **Inferior** (or **caudal**) describes a position below or lower than another part of the body proper; near or toward the tail (in humans, the coccyx, or lowest part of the spinal column). The pelvis is inferior to the abdomen.
- **Lateral** describes the side or direction toward the side of the body. The thumb (pollex) is lateral to the digits.
- **Medial** describes the middle or direction toward the middle of the body. The hallux is the medial toe.
- **Proximal** describes a position in a limb that is nearer to the point of attachment or the trunk of the body. The brachium is proximal to the antebrachium.
- **Distal** describes a position in a limb that is farther from the point of attachment or the trunk of the body. The crus is distal to the femur.
- **Superficial** describes a position closer to the surface of the body. The skin is superficial to the bones.
- **Deep** describes a position farther from the surface of the body. The brain is deep to the skull.



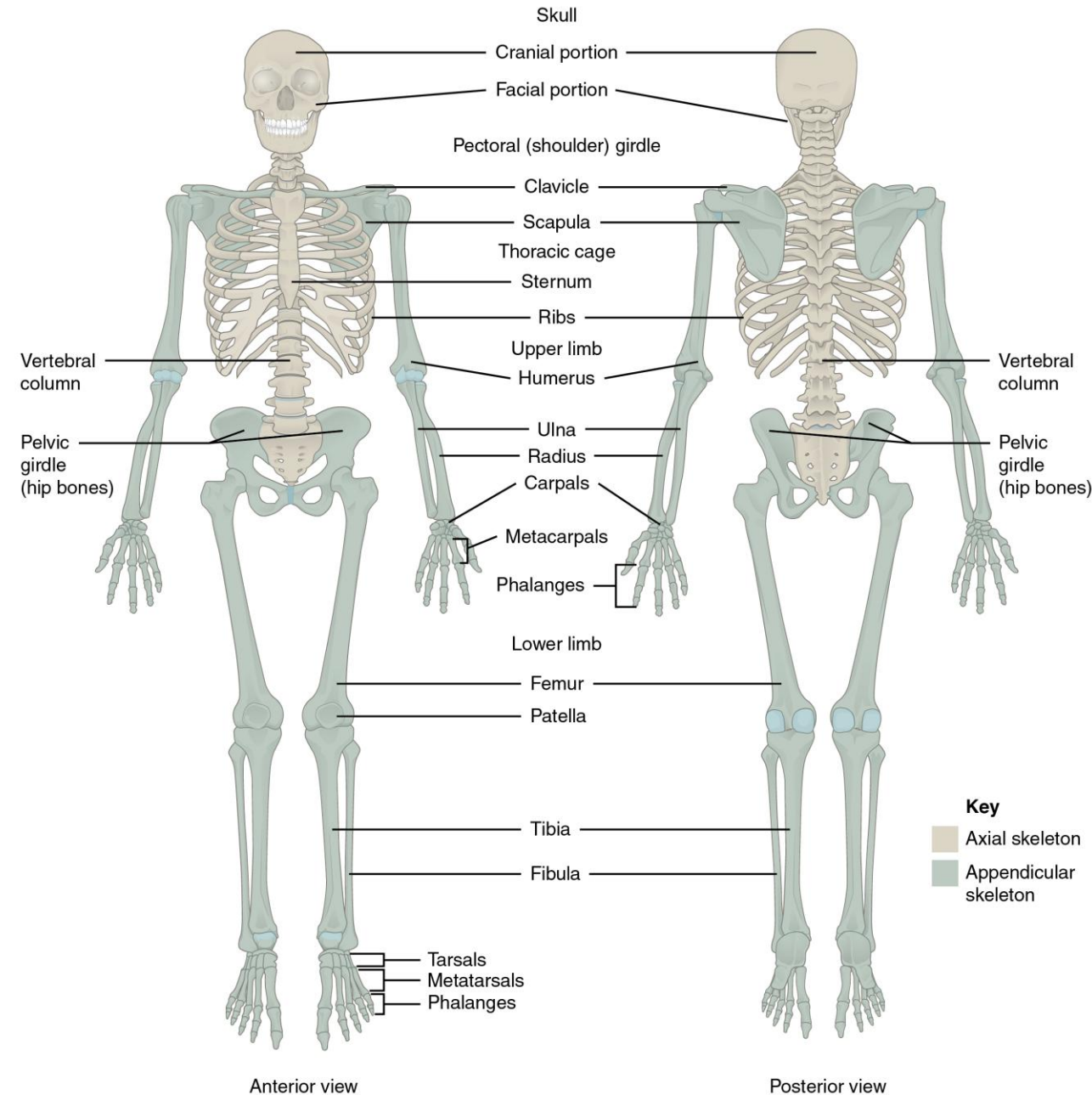
Body Planes

- The **sagittal plane** is the plane that divides the body or an organ vertically into **right and left** sides. If this vertical plane runs directly down the middle of the body, it is called the midsagittal or median plane. If it divides the body into unequal right and left sides, it is called a parasagittal plane or less commonly a longitudinal section.
- The **frontal plane** is the plane that divides the body or an organ into an anterior (**front**) portion and a posterior (**rear**) portion. The frontal plane is often referred to as a **coronal plane**. (“Corona” is Latin for “crown.”)
- The **transverse plane** is the plane that divides the body or organ horizontally into **upper and lower** portions. Transverse planes produce images referred to as **cross sections**.



Axial skeleton

- The **axial skeleton** forms the vertical, central axis of the body and includes all bones of the head, neck, chest, and back.
- It serves to protect the brain, spinal cord, heart, and lungs. It also serves as the attachment site for muscles that move the head, neck, and back, and for muscles that act across the shoulder and hip joints to move their corresponding limbs.
- The axial skeleton of the adult consists of 80 bones, including the **skull**, the **vertebral column**, and the **thoracic cage**.
 - The skull is formed by 22 bones.
 - The vertebral column consists of 24 bones, each called a **vertebra**, plus the **sacrum** and **coccyx**.
 - The thoracic cage includes the 12 pairs of **ribs**, and the **sternum**, the flattened bone of the anterior chest.

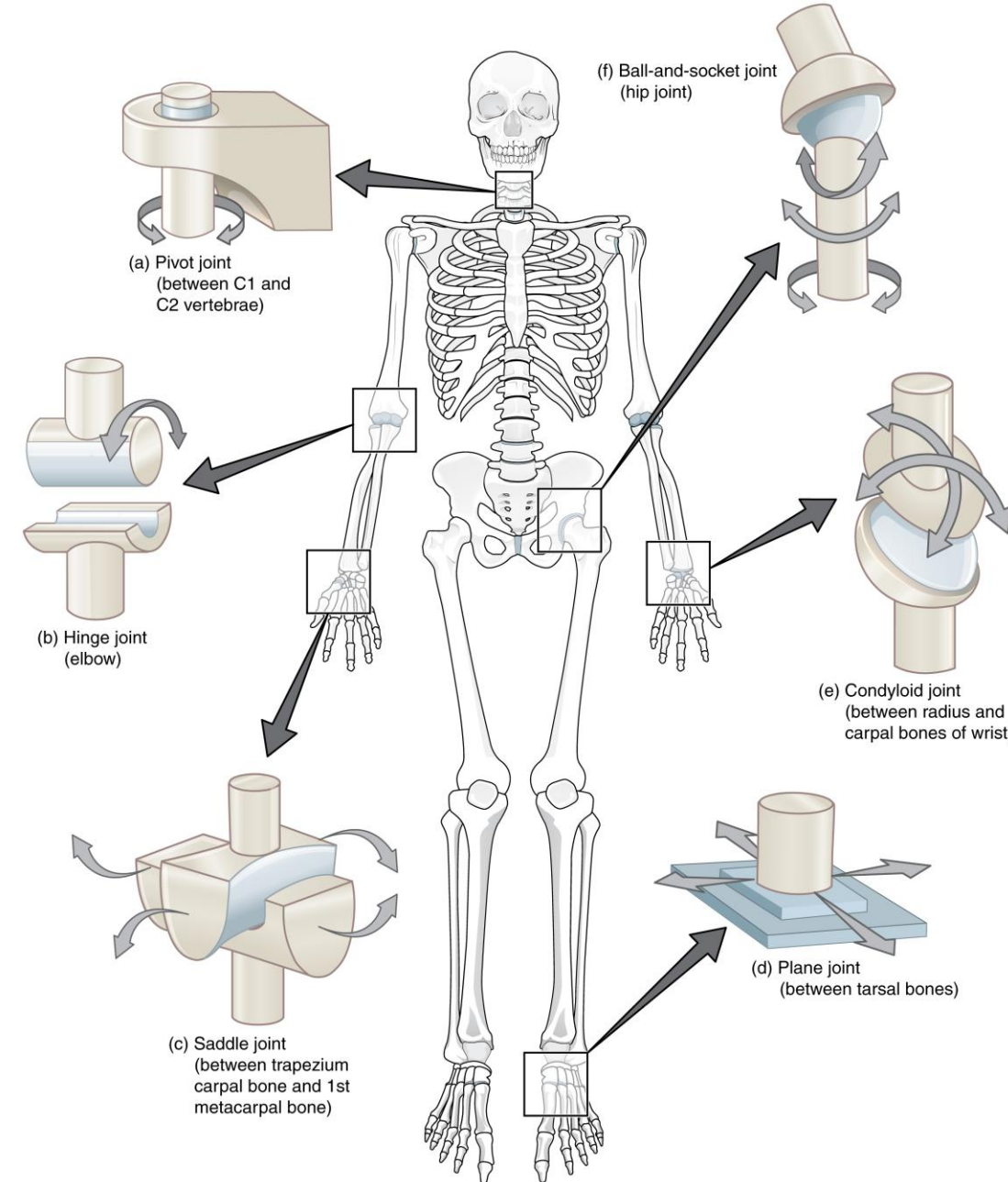


Joints

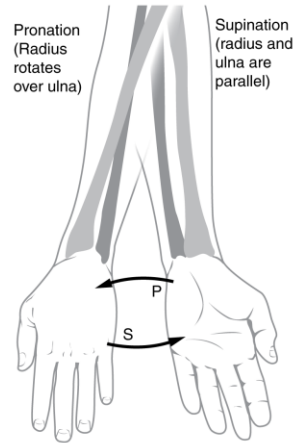
- Joints are divided into three categories, based on the number of axes of motion provided by each.
- An axis in anatomy is described as the movements in reference to the three anatomical planes: transverse, frontal, and sagittal.
- Joints are classified as:
 - **Uniaxial joints:** they only allow for a motion in a single plane (around a single axis)
 - The elbow joint, which only allows for bending or straightening, is an example of a uniaxial joint
 - **Biaxial joints:** they allow for motions within two planes
 - An example of a biaxial joint is a metacarpophalangeal joint (knuckle joint) of the hand. The joint allows for movement along one axis to produce bending or straightening of the finger, and movement along a second axis, which allows for spreading of the fingers away from each other and bringing them together
 - **Multiaxial joints** (polyaxial or triaxial joints): they allow for the several directions of movement, for movement along three axes
 - The shoulder and hip joints are multiaxial joints. They allow the upper or lower limb to move in an anterior-posterior direction and a medial-lateral direction. In addition, the limb can also be rotated around its long axis. This third movement results in rotation of the limb so that its anterior surface is moved either toward or away from the midline of the body.

Types of Joints

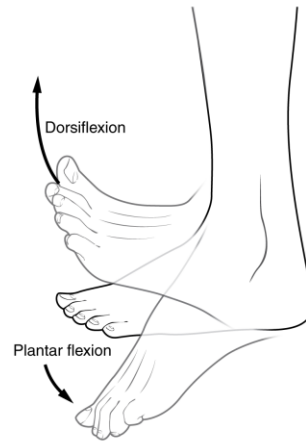
- (a) **Pivot joints** allow for rotation around an axis, such as between the first and second cervical vertebrae, which allows for side-to-side rotation of the head.
- (b) The **hinge joint** of the elbow works like a door hinge.
- (c) The articulation between the trapezium carpal bone and the first metacarpal bone at the base of the thumb is a **saddle joint**.
- (d) **Plane joints**, such as those between the tarsal bones of the foot, allow for limited gliding movements between bones.
- (e) The radiocarpal joint of the wrist is a **condyloid joint**.
- (f) The hip and shoulder joints are the only **ball-and-socket joints** of the body.



Types of movements



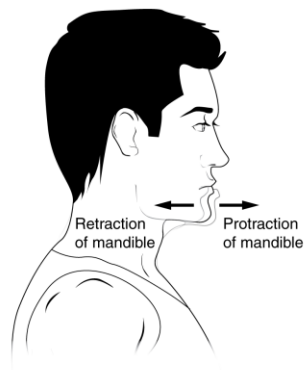
(g) Pronation (P) and supination (S)



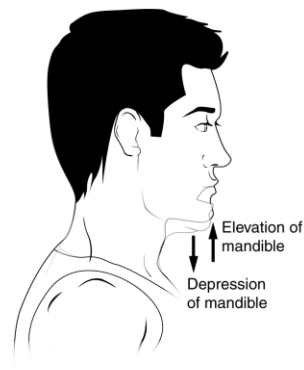
(h) Dorsiflexion and plantar flexion



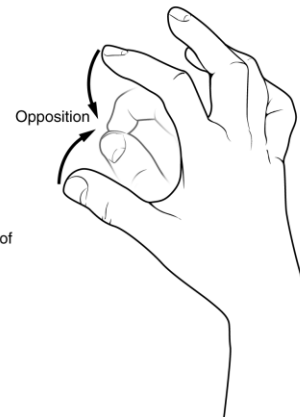
(i) Inversion and eversion



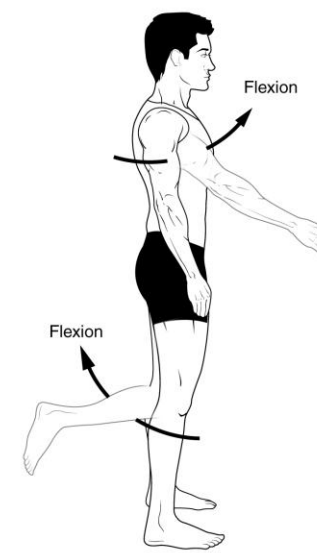
(j) Protraction and retraction



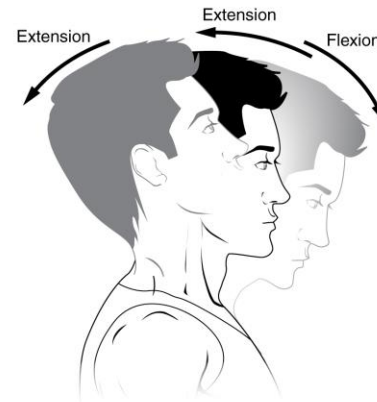
(k) Elevation and depression



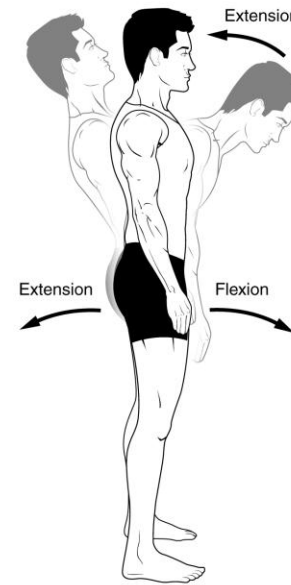
(l) Opposition



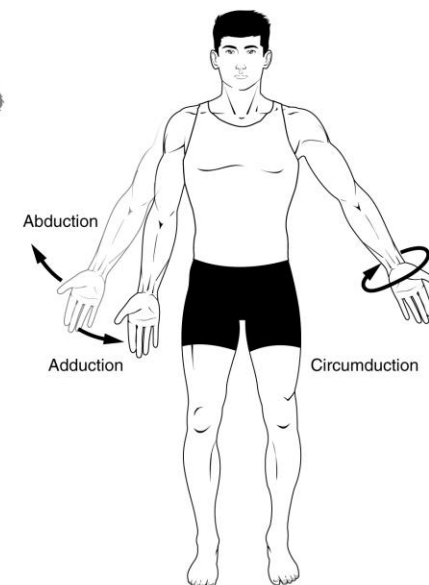
(a) and (b) Angular movements: flexion and extension at the shoulder and knees



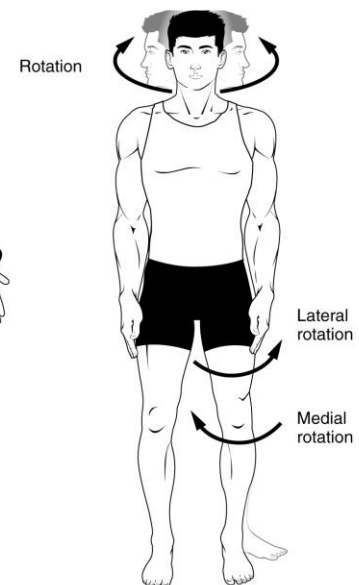
(c) Angular movements: flexion and extension of the neck



(d) Angular movements: flexion and extension of the vertebral column



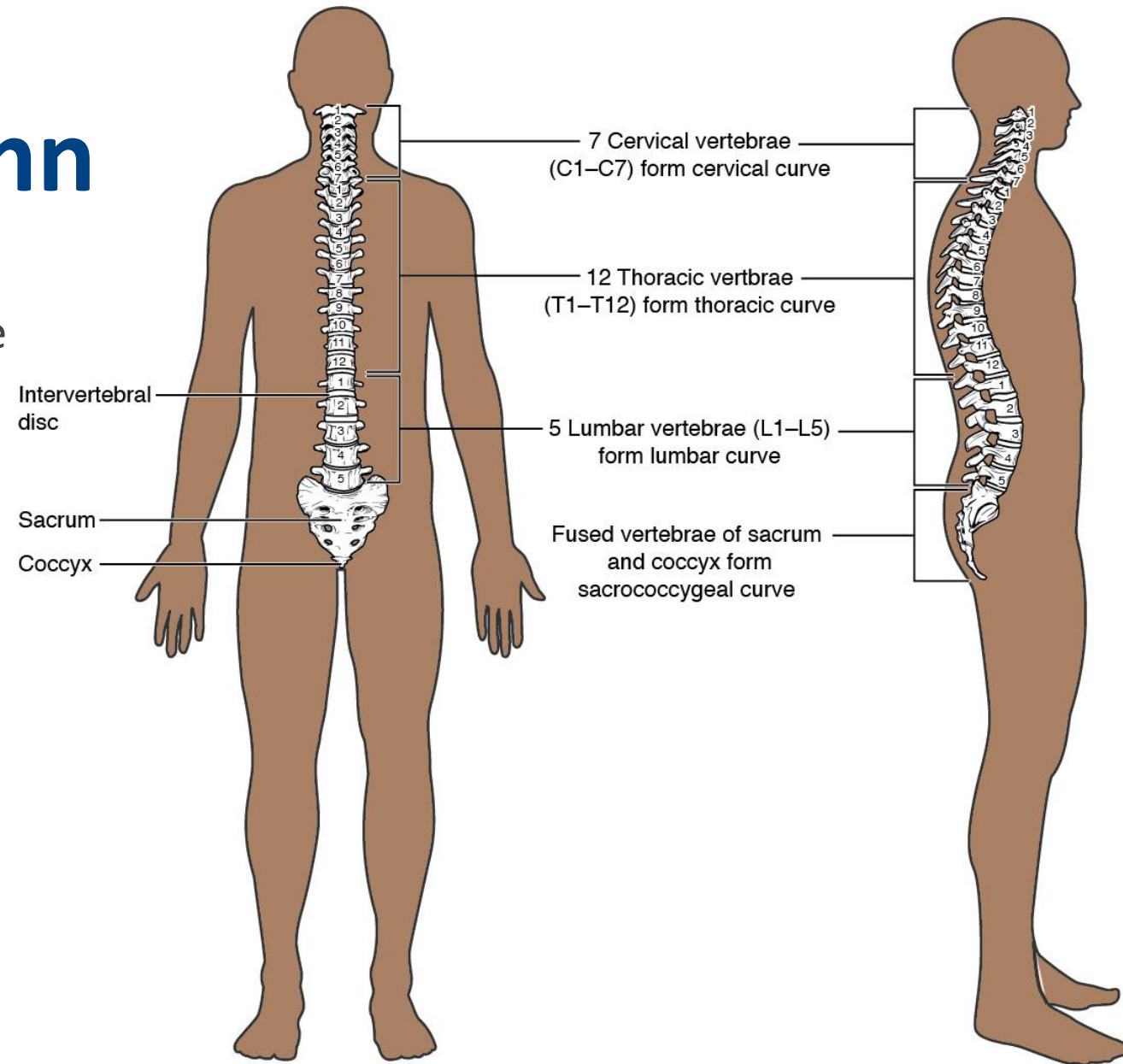
(e) Angular movements: abduction, adduction, and circumduction of the upper limb at the shoulder



(f) Rotation of the head, neck, and lower limb

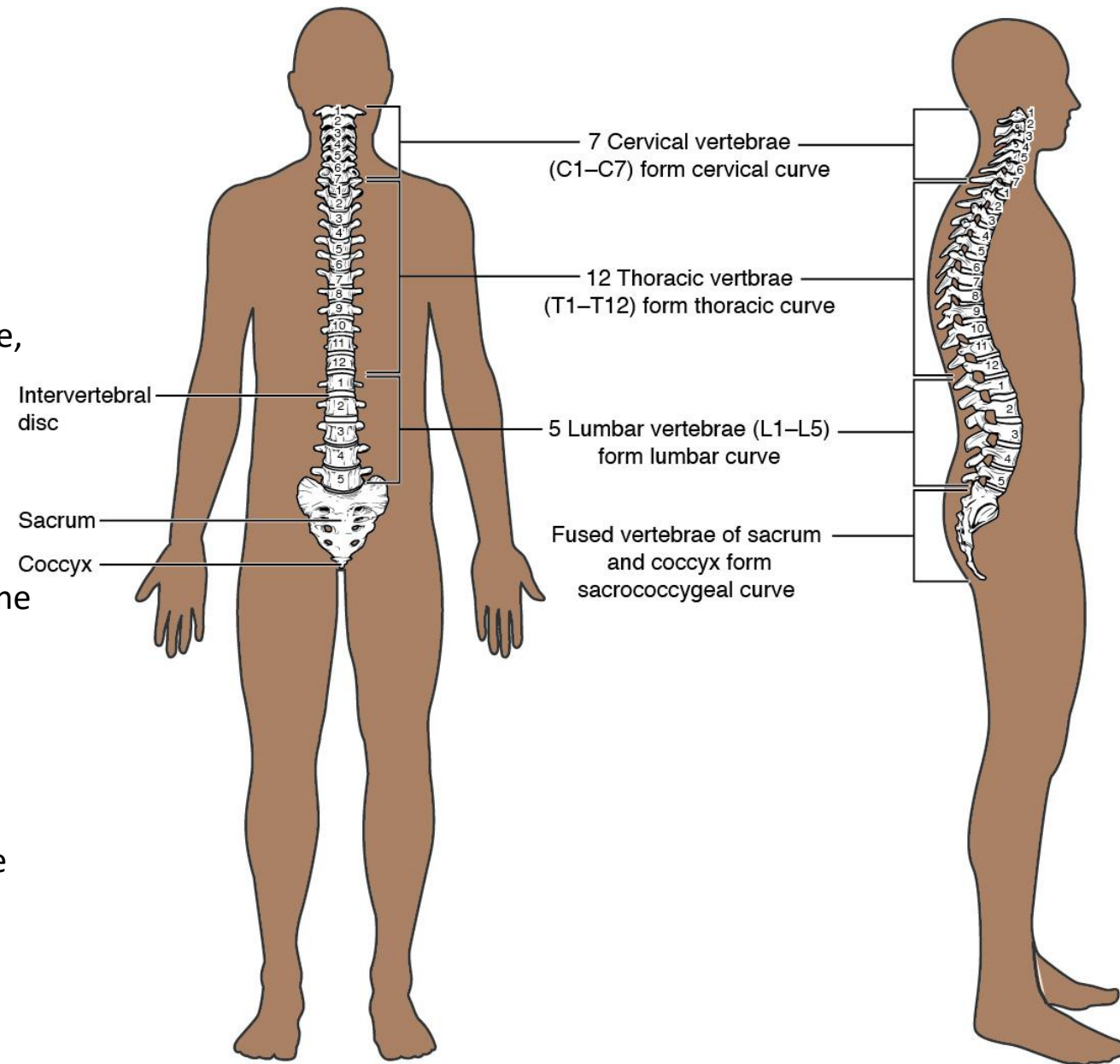
The vertebral column

- Also known as the spinal column or spine
- It consists of a sequence of vertebrae (singular = vertebra), each of which is separated and united by an **intervertebral disc**. Together, the vertebrae and intervertebral discs form the vertebral column.
- It is a flexible column that supports the head, neck, and body and allows for their movements.
- It also protects the spinal cord, which passes down the back through openings in the vertebrae.



Regions of the vertebral column

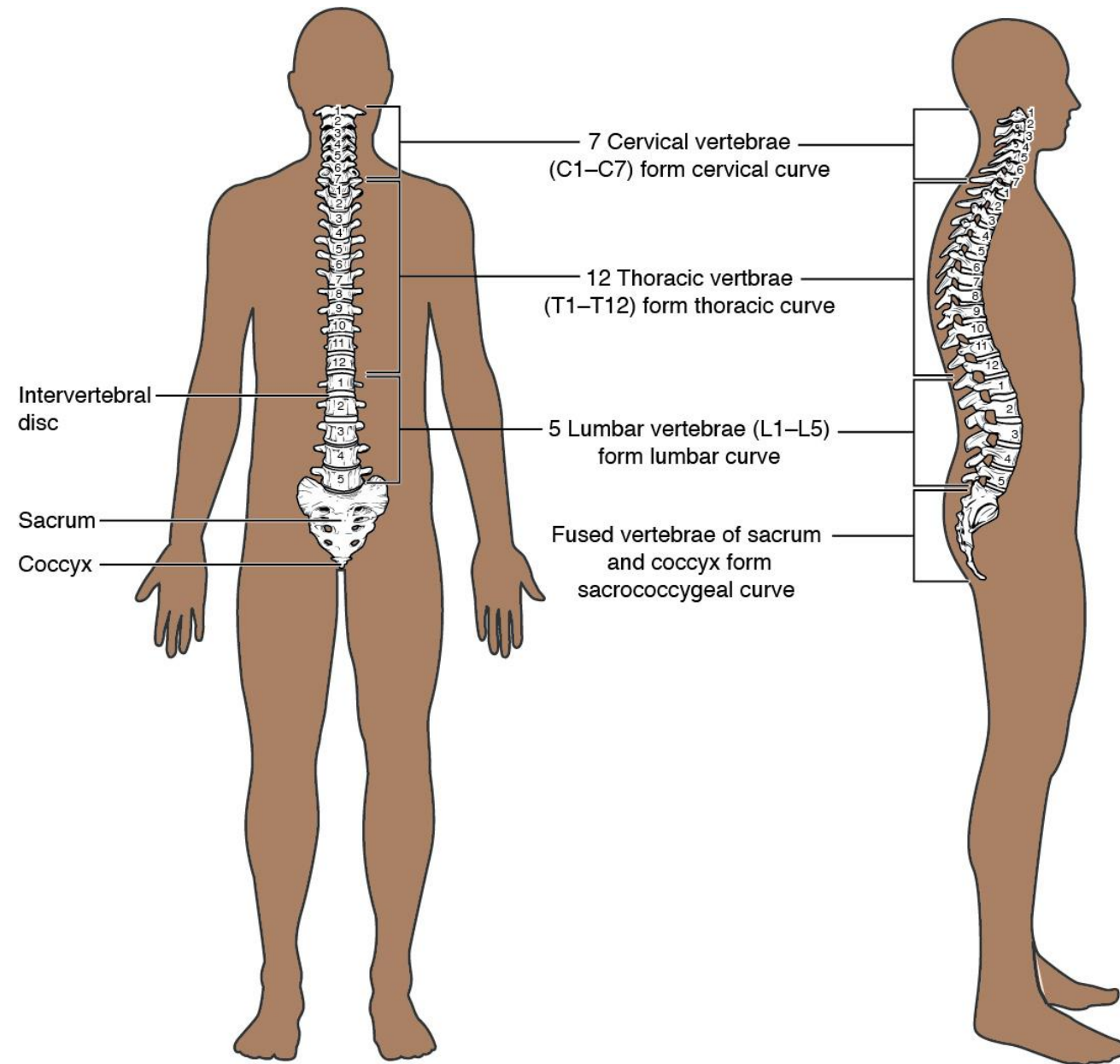
- The vertebral column originally develops as a series of 33 vertebrae, but this number is eventually reduced to 24 vertebrae, plus the sacrum and coccyx.
- The vertebral column is subdivided into **five regions**, with the vertebrae in each area named for that region and numbered in **descending** order.
- In the neck, there are **7 cervical vertebrae**, each designated with the letter “C” followed by its number. Superiorly, the C1 vertebra articulates (forms a joint) with the occipital condyles of the skull. Inferiorly, C1 articulates with the C2 vertebra, and so on.
- Below these are the **12 thoracic vertebrae**, designated **T1–T12**.
- The lower back contains the **L1–L5 lumbar vertebrae**.
- The single **sacrum**, which is also part of the pelvis, is formed by the **fusion of five sacral vertebrae**.
- The **coccyx**, or tailbone, results from the **fusion of four small coccygeal vertebrae**. However, the sacral and coccygeal fusions do not start until age 20 and are not completed until middle age.



An interesting anatomical fact is that almost all mammals have seven cervical vertebrae, regardless of body size. This means that there are large variations in the size of cervical vertebrae, ranging from the very small cervical vertebrae of a shrew to the greatly elongated vertebrae in the neck of a giraffe. In a full-grown giraffe, each cervical vertebra is 11 inches

Curvatures of the vertebral column

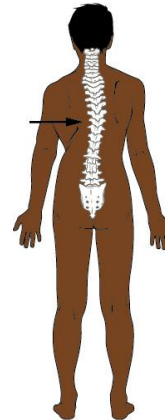
- The adult vertebral column does not form a straight line, but instead has **four curvatures** along its length.
- These curves increase the vertebral column's strength, flexibility, and ability to absorb shock. When the load on the spine is increased, by carrying a heavy backpack for example, the curvatures increase in depth (become more curved) to accommodate the extra weight. They then spring back when the weight is removed.
- The four adult curvatures are classified as either **primary** or **secondary** curvatures. Primary curves are retained from the original fetal curvature, while secondary curvatures develop after birth.
- In the adult, fetal curvature is retained in two regions of the vertebral column as the **thoracic curve**, which involves the thoracic vertebrae, and the **sacroccocygeal curve**, formed by the sacrum and coccyx.
- Secondary curves are concave posteriorly, opposite in direction to the original fetal curvature. The **cervical curve** of the neck region develops as the infant begins to hold their head upright when sitting. Later, as the child begins to stand and then to walk, the **lumbar curve** of the lower back develops. In adults, the lumbar curve is generally deeper in females.



Disorders of the vertebral column

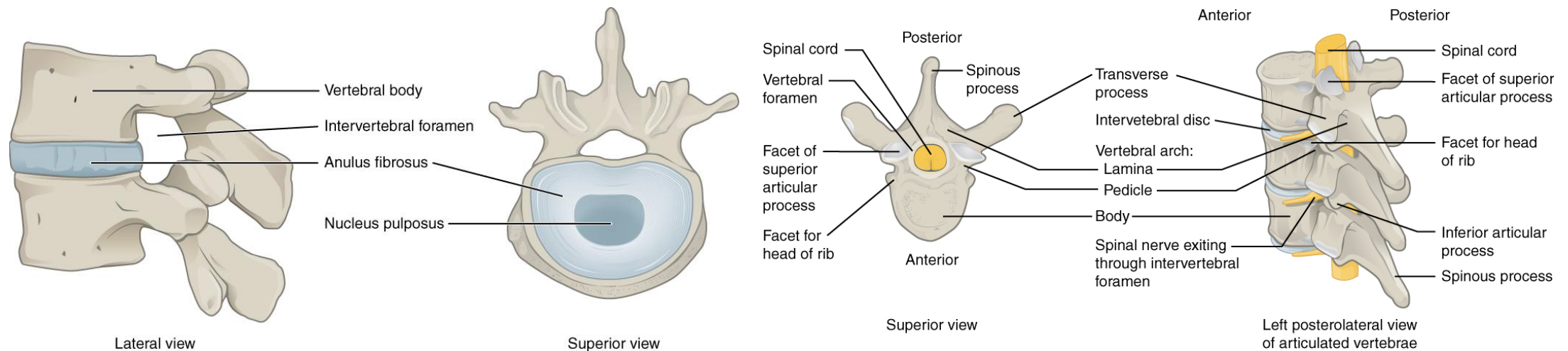
Developmental anomalies, pathological changes, or obesity can enhance the normal vertebral column curves, resulting in the development of abnormal or excessive curvatures.

- **Kyphosis**, also referred to as humpback or hunchback, is an **excessive posterior curvature of the thoracic region**. This can develop when osteoporosis causes weakening and erosion of the anterior portions of the upper thoracic vertebrae, resulting in their gradual collapse.
- **Lordosis**, or swayback, is an **excessive anterior curvature of the lumbar region** and is most commonly associated with obesity or late pregnancy. The accumulation of body weight in the abdominal region results in an anterior shift in the line of gravity that carries the weight of the body. This causes an anterior tilt of the pelvis and a pronounced enhancement of the lumbar curve.
- **Scoliosis** is an abnormal, **lateral curvature**, accompanied by twisting of the vertebral column. Compensatory curves may also develop in other areas of the vertebral column to help maintain the head positioned over the feet. Scoliosis is the most common vertebral abnormality among girls. The cause is usually unknown, but it may result from weakness of the back muscles, defects such as differential growth rates in the right and left sides of the vertebral column, or differences in the length of the lower limbs. When present, scoliosis tends to get worse during adolescent growth spurts. Although most individuals do not require treatment, a back brace may be recommended for growing children. In extreme cases, surgery may be required.



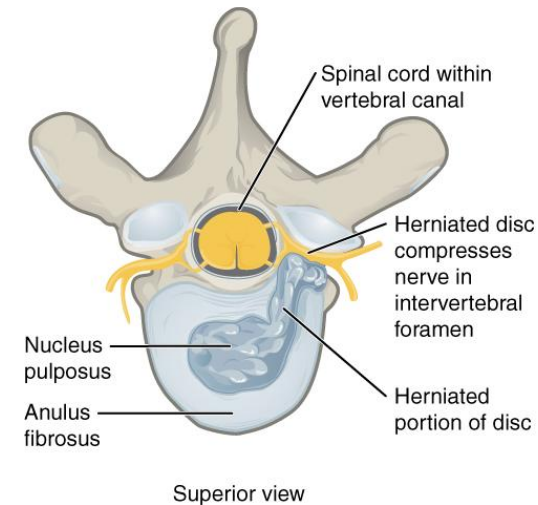
Structure of a Vertebra

- Within the different regions of the vertebral column, vertebrae vary in size and shape, but they all follow a similar structural pattern. A typical vertebra will consist of a **body**, a **vertebral arch**, and **seven processes**.
- The **body** is the anterior portion of each vertebra and is the part that supports the body weight. Because of this, the vertebral bodies progressively increase in size and thickness going down the vertebral column. The bodies of adjacent vertebrae are separated and strongly united by an **intervertebral disc**.
- The large opening between the vertebral arch and body is the **vertebral foramen**, which contains the **spinal cord**. In the intact vertebral column, the vertebral foramina of all of the vertebrae align to form the **vertebral (spinal) canal**, which serves as the bony protection and passageway for the spinal cord down the back. When the vertebrae are aligned together in the vertebral column, notches in the margins of the pedicles of adjacent vertebrae together form an **intervertebral foramen**, the opening through which a spinal nerve exits from the vertebral column.



Intervertebral discs

- An **intervertebral disc** is a fibrocartilaginous pad that fills the gap between adjacent vertebral bodies.
- Each disc is anchored to the bodies of its adjacent vertebrae, thus strongly uniting these.
- The discs also provide padding between vertebrae during weight bearing. Because of this, intervertebral discs are thin in the cervical region and thickest in the lumbar region, which carries the most body weight.
- The **nucleus pulposus** consists of a softer, more gel-like material. It has a high water content that serves to resist compression and thus is important for weight bearing.
- The gel-like nature of the nucleus pulposus also allows the intervertebral disc **to change shape** as one vertebra rocks side to side or forward and back in relation to its neighbors during movements of the vertebral column. Thus, bending forward causes compression of the anterior portion of the disc but expansion of the posterior disc.
- If the posterior anulus fibrosus is weakened due to injury or increasing age, the pressure exerted on the disc when bending forward and lifting a heavy object can cause the nucleus pulposus to protrude posteriorly through the anulus fibrosus, resulting in a **herniated disc** (“ruptured” or “slipped” disc). The posterior bulging of the nucleus pulposus can cause **compression of a spinal nerve** at the point where it exits through the intervertebral foramen, with resulting pain and/or muscle weakness in those body regions supplied by that nerve.
- The most common sites for disc herniation are the L4/L5 or L5/S1 intervertebral discs, which can cause sciatica, a widespread pain that radiates from the lower back down the thigh and into the leg. Similar injuries of the C5/C6 or C6/C7 intervertebral discs, following forcible hyperflexion of the neck from a collision accident or football injury, can produce pain in the neck, shoulder, and upper limb.

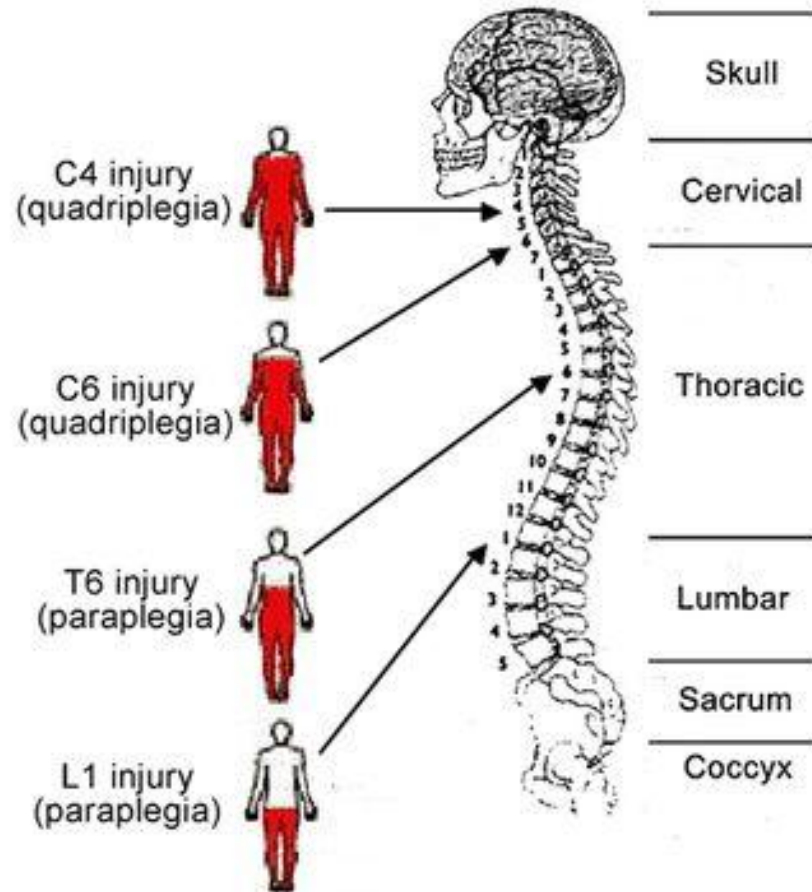


Spinal cord injuries (SCI)



SCI symptoms

Level of Injury and Extent of Paralysis



VERTEBRAL LEVEL	NERVE ROOT*	INNERVATION	POSSIBLE SYMPTOMS
C1	C1	Intracranial Blood Vessels	Headaches + Migraine Headaches
C2	C2	• Eyes • Lacrimal Gland	• Dizziness • Sinus Problems
C3	C3	• Parotid Gland • Scalp	• Allergies • Head Colds • Fatigue
C4	C4	• Base of Skull • Neck Muscles • Diaphragm	• Vision Problems • Runny Nose • Sore Throat • Stiff Neck
C5	C5	• Neck Muscles • Shoulders	• Cough • Croup • Arm Pain
C6	C6	• Elbows • Arms • Wrists	• Hand and Finger Numbness or Tingling • Asthma • Heart Conditions • High Blood Pressure
C7	C7	• Hands • Fingers • Esophagus • Heart • Lungs • Chest	
T1	T1	Arms • Esophagus	Wrist, Hand and Finger Numbness or Pain • Middle Back Pain • Congestion • Difficulty Breathing • Asthma • High Blood Pressure • Heart Conditions
T2	T2	• Heart • Lungs • Chest	
T3	T3	• Larynx • Trachea	
T4	T4		
T5	T5	Gallbladder • Liver	
T6	T6	• Diaphragm • Stomach	• Bronchitis • Pneumonia
T7	T7	• Pancreas • Spleen	• Gallbladder Conditions
T8	T8	• Kidneys • Small Intestine	• Jaundice • Liver Conditions
T9	T9	• Appendix • Adrenals	• Stomach Problems • Ulcers
T10	T10		
T11	T11	Small Intestines • Colon • Uterus	• Gastritis • Kidney Problems
T12	T12	Uterus • Colon • Buttocks	
L1	L1	Large Intestines	Constipation • Colitis • Diarrhea
L2	L2	• Buttocks • Groin	• Gas Pain • Irritable Bowel
L3	L3	• Reproductive Organs	• Bladder Problems • Menstrual Problems • Low Back Pain
L4	L4	• Colon • Thighs • Knees	• Pain or Numbness in Legs
L5	L5	• Legs • Feet	
SACRAL	SACRAL	Buttocks • Reproductive Organs • Bladder • Prostate Gland • Legs • Ankles • Feet • Toes	Constipation • Diarrhea • Bladder Problems • Menstrual Problems • Lower Back Pain • Pain or Numbness in Legs

SCI symptoms

SCIs may lead to 4 different clinical conditions:

- **Paraplegia:** sensory-motor syndrome with paralysis of lower limbs, due to SCI of **thoracic or lumbar** spinal cord tract
- **Tetraplegia:** more severe sensory-motor syndrome that adds to paraplegia impairs a paralysis of upper limbs and neurovegetative disorders. It is due to SCI of the **cervical** spinal cord tract.
- **Paraparesis:** incomplete paralysis of lower limbs
- **Tetraparesis:** incomplete paralysis of all limbs

	Complete	Incomplete
Lower limbs	Paraplegia	Paraparesis
Upper and lower limbs	Tetraplegia	Tetraparesis



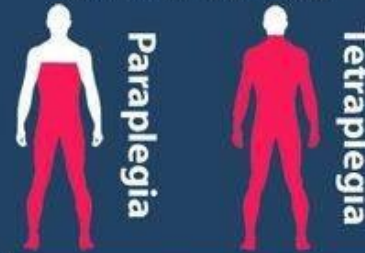
Spinal Cord Injury Infographic Types of Spinal Cord Injury

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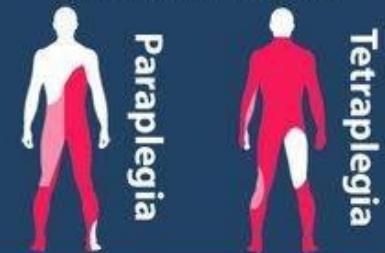
Complete Spinal Cord Injury

Complete loss of motor and sensory function below the spinal cord injury.



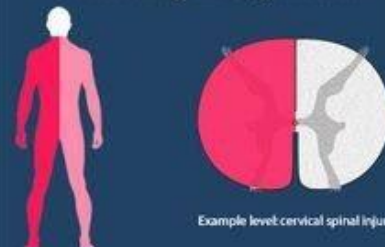
Incomplete Spinal Cord Injury

Partial random preservation of motor or sensory function below the spinal cord injury.



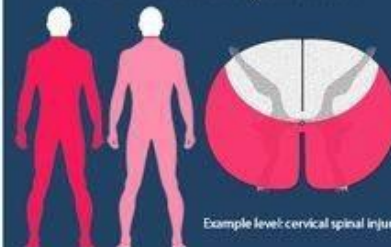
Common Types of Incomplete Spinal Cord Injuries

Brown-Séquard Syndrome



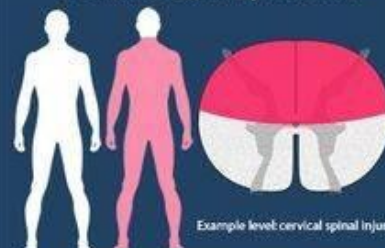
Below injury level, motor weakness or paralysis on one side of the body (hemiparaplegia). Loss of sensation on the opposite side (hemi-anesthesia).

Anterior Cord Syndrome



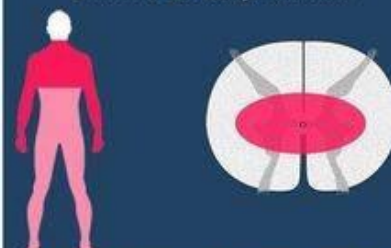
Below injury level, motor paralysis and loss of pain and temperature sensation. Proprioception (position sense), touch and vibratory sensation preserved.

Posterior Cord Syndrome



Below injury level, motor function preserved. Loss of sensory function: pressure, stretch, and proprioception (position sense).

Central Cord Syndrome



Results from cervical spinal injuries. Greater motor impairment in upper body compared to lower body. Variable sensory loss below the level of injury.

Cervical Nerves:

Diaphragm
Deltoids
Biceps
Wrist extensors
Rotates arm
Triceps
Bends fingers

C1
C2
C3
C4
C5
C6
C7

Thoracic Nerves:

Spread fingers
Chest muscles
Abdominal muscles
Muscles in the back

T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12

Lumbar Nerves:

Hip muscles
Thigh muscles
Knee Muscles
Foot muscles

L1
L2
L3
L4
L5

Sacral Nerves:

Bladder and bowel
Sexual function

S1
S2
S3
S4
S5

Cervical

Thoracic

Lumbar

Sacral

Sources: www.apparelyzed.com
www.wikipedia.org
sci.rutgers.edu

Key:

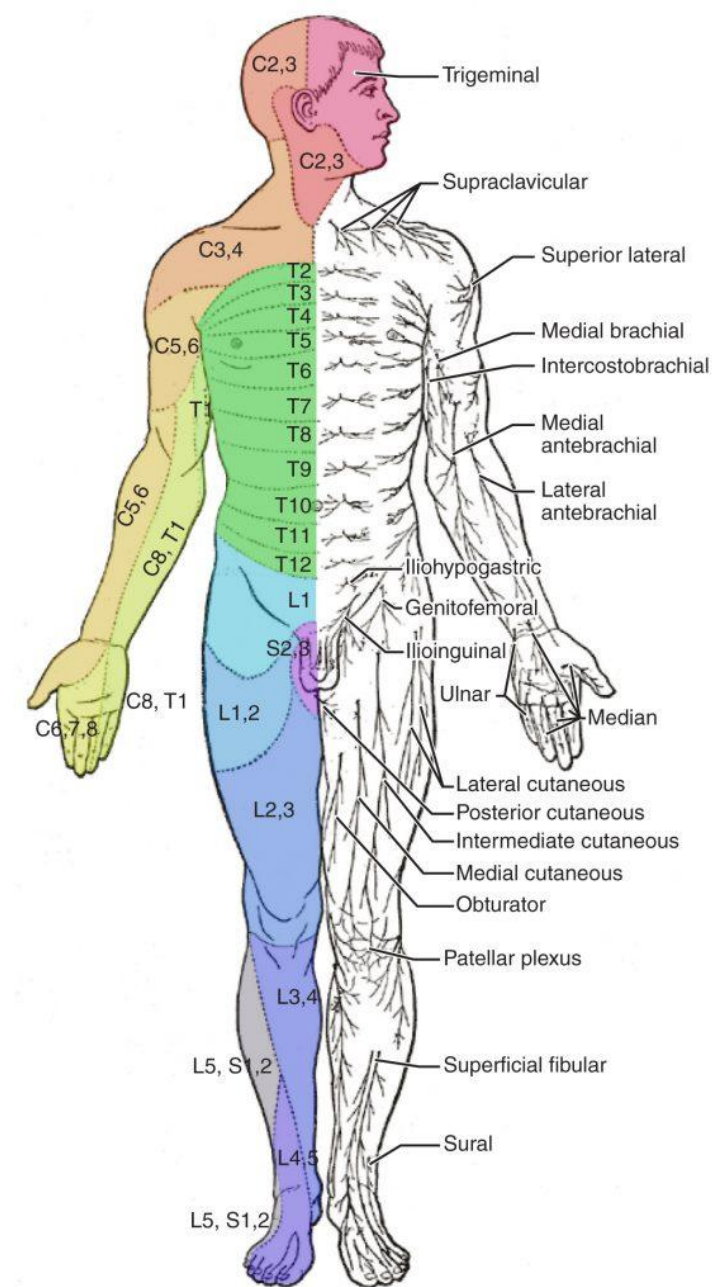
■ Normal Function
■ Impaired Motor Function
■ Impaired Sensory Function

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Dermatomes and myotomes

- **Dermatomes:**
skin area innervated by the **dorsal (sensory)** root of a single spinal nerve
- **Myotomes:**
skin area innervated by the **ventral (motor)** root of a single spinal nerve



ASIA scale

STANDARD NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY

ASIA

MOTOR

KEY MUSCLES

	R	L
C2		
C3		
C4		
C5		
C6		
C7		
C8		
T1		
T2		
T3		
T4		
T5		
T6		
T7		
T8		
T9		
T10		
T11		
T12		
L1		
L2		
L3		
L4		
L5		
S1		
S2		
S3		
S4-5		

0 = total paralysis
 1 = palpable or visible contraction
 2 = active movement, gravity eliminated
 3 = active movement, against gravity
 4 = active movement, against some resistance
 5 = active movement, against full resistance
 NT = not testable

Elbow flexors
 Wrist extensors
 Elbow extensors
 Finger flexors (distal phalanx of middle finger)
 Finger abductors (little finger)

Hip flexors
 Knee extensors
 Ankle dorsiflexors
 Long toe extensors
 Ankle plantar flexors

☐ Voluntary anal contraction (Yes/No)

TOTALS + = **MOTOR SCORE**
 (MAXIMUM) (50) (50) (100)

SENSORY

KEY SENSORY POINTS

0 = absent
 1 = impaired
 2 = normal
 NT = not testable

LIGHT TOUCH

	R	L
C2		
C3		
C4		
C5		
C6		
C7		
C8		
T1		
T2		
T3		
T4		
T5		
T6		
T7		
T8		
T9		
T10		
T11		
T12		
L1		
L2		
L3		
L4		
L5		
S1		
S2		
S3		
S4-5		

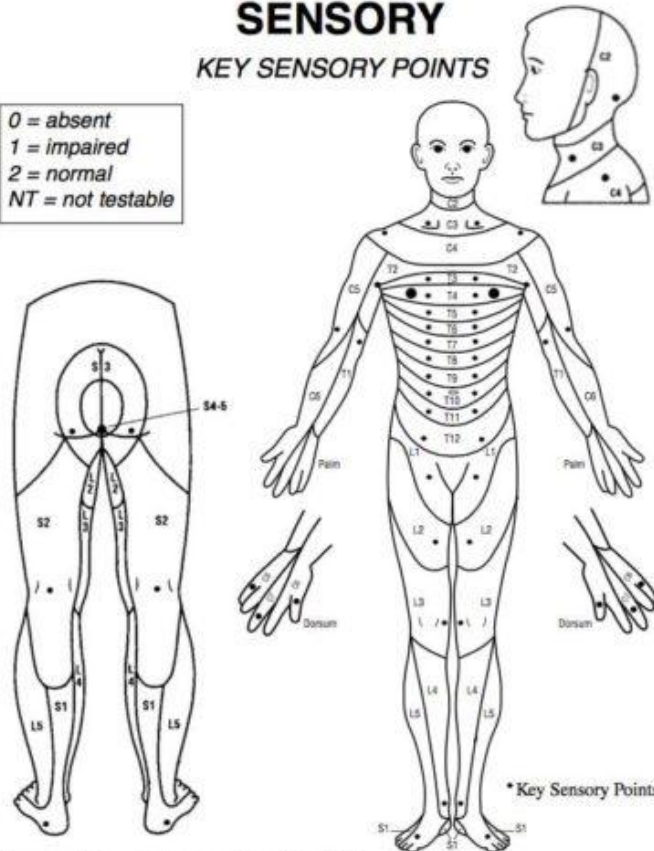
PIN PRICK

	R	L
C2		
C3		
C4		
C5		
C6		
C7		
C8		
T1		
T2		
T3		
T4		
T5		
T6		
T7		
T8		
T9		
T10		
T11		
T12		
L1		
L2		
L3		
L4		
L5		
S1		
S2		
S3		
S4-5		

Any anal sensation (Yes/No)

TOTALS + = **PIN PRICK SCORE** (max: 112)
 (MAXIMUM) (56) (56) (56) (56)

TOTALS + = **LIGHT TOUCH SCORE** (max: 112)
 (MAXIMUM) (56) (56) (56) (56)



NEUROLOGICAL LEVEL The most caudal segment with normal function	R	L	COMPLETE OR INCOMPLETE? Incomplete = Any sensory or motor function in S4-S5	ZONE OF PARTIAL PRESERVATION Caudal extent of partially innervated segments	R	L
	SENSORY	MOTOR			SENSORY	MOTOR

ASIA IMPAIRMENT SCALE

ASIA scale



https://www.youtube.com/watch?v=hO9hADODTw8&ab_channel=UCTeachOrtho

Assessment scales

- Measurement of **impairment of body functions or structure** (previously impairment*)
- Measurement of **activity limitation** (previously disability*)
- Measurement of **participation restriction** (previously handicap*)
- Measurement of **quality of life** or health status

* ICDH: *International Classification of Impairment, Disability and Handicap*

Assesment scales for impairment

- Quantify the effect of diseases on anatomical structures or functional systems of human organism
- Most common scales:
 - Motricity Index
 - ASIA (measures the level of spinal cord injuries)
 - NIH (measures the neurological status of post-stroke patients)
 - Ashworth Scale (measures the level of spasticity of a muscular group)

Assessment scales for disability

- Assess the activities of daily living (ADL) or the instrumental activities of daily living (IADL*)
- The ADL tools measure personal activities like:
 - dressing up
 - eating
 - bathing
 - personal care
- The IADL tools measure the patient's capabilities of:
 - adapt to the environment
 - house keeping
 - shopping
- Most common scales:
 - Barthel Index
 - FIM
 - PULSES Profile

*Instrumental activities of daily living (IADL) are those activities that allow an individual to live independently in a community. Although not necessary for functional living, the ability to perform IADLs can significantly improve the quality of life. The major domains of IADLs include cooking, cleaning, transportation, laundry, and managing finances.

Barthel Index (1965)

The Barthel Index assesses ten common ADLs, including basic mobility:

- Feeding
- Bathing
- Grooming
- Dressing
- Bowel
- Bladder
- Toilet use
- Transfers bed-to-chair-and-back
- Mobility on level surfaces
- Stair negotiation

Barthel Index

Typically, the assessment takes only **minutes** to complete and can be a part of a rehabilitation evaluation.

- The Barthel Index measures the **degree of assistance** required by an individual on ten mobility and self-care ADL items
- The time taken and physical assistance required to perform each item are used to determine the assigned **value** of each item
- The ten items are scored with a **number of points**, and then a final score is calculated by summing the points awarded to each functional skill. This allows the examiner to measure a patient's functional disability by quantifying their performance.

Barthel Index scoring

- Each item is scored based on whether or not the individual can perform a task or activity independently, with assistance or if they are fully dependent:
 - 0 = unable**
 - 1 = needs assistance/help**
 - 2 = independent**
- The ten items are summed and x 5 to get a total score out of 100.
- Proposed guidelines for interpreting Barthel scores are as follows:
 - scores of **0-20** indicate “**total**” dependency
 - scores of **21-60** indicate “**severe**” dependency
 - scores of **61-90** indicate “**moderate**” dependency
 - scores of **91-99** indicate “**slight**” dependency

<https://www.sralab.org/sites/default/files/2017-07/barthel.pdf>

THE BARTHEL INDEX

Patient Name: _____
Rater Name: _____
Date: _____

Activity	Score
FEEDING 0 = unable 5 = needs help cutting, spreading butter, etc., or requires modified diet 10 = independent	_____
BATHING 0 = dependent 5 = independent (or in shower)	_____
GROOMING 0 = needs to help with personal care 5 = independent face/hair/teeth/shaving (implements provided)	_____
DRESSING 0 = dependent 5 = needs help but can do about half unaided 10 = independent (including buttons, zips, laces, etc.)	_____
BOWELS 0 = incontinent (or needs to be given enemas) 5 = occasional accident 10 = continent	_____
BLADDER 0 = incontinent, or catheterized and unable to manage alone 5 = occasional accident 10 = continent	_____
TOILET USE 0 = dependent 5 = needs some help, but can do something alone 10 = independent (on and off, dressing, wiping)	_____
TRANSFERS (BED TO CHAIR AND BACK) 0 = unable, no sitting balance 5 = major help (one or two people, physical), can sit 10 = minor help (verbal or physical) 15 = independent	_____
MOBILITY (ON LEVEL SURFACES) 0 = immobile or < 50 yards 5 = wheelchair independent, including corners, > 50 yards 10 = walks with help of one person (verbal or physical) > 50 yards 15 = independent (but may use any aid; for example, stick) > 50 yards	_____
STAIRS 0 = unable 5 = needs help (verbal, physical, carrying aid) 10 = independent	_____
TOTAL (0-100):	_____

Barthel Index scoring

- Scores can be assigned either via direct assessment / **observation** or from reliable **interviews** with the patient, family, or staff. Research also suggests that the examiner's "common sense" and clinical experience can be used to assign scores
- **The higher the score, the more independent the patient is** in completing the measured ADLs. Higher scores also indicate the patient is more likely to return home, with varying degrees of assistance, following hospital discharge
- **The lower the score, the more dependent the patient is** with ADL completion, and the more skilled care will be required at discharge
- most studies use a score of 60/61 (moderate dependency) as a cutting point

THE BARTHEL INDEX

Patient Name: _____
Rater Name: _____
Date: _____

Activity	Score
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STAIRS 0 = unable 5 = needs help (verbal, physical, carrying aid) 10 = independent	_____

TOTAL (0-100): _____

Assessment scales for handicap

- Assess psychological, social and economical aspects of the patient
- Most common scales:
 - CHART Questionnaire (Craig Handicap Assessment & Reporting Technique)
 - CIQ (Community Integration Questionnaire)

CHART Questionnaire (Craig Handicap Assessment & Reporting Technique)

- Designed to assess how people with disabilities function as active members of their communities
- 6 domains:
 - Physical independence
 - Cognitive independence
 - Mobility
 - Occupation
 - Social integration
 - Economic self-sufficiency
- 32 items in total
- Scores on each subscale (domain) range from 0-100 with total CHART score ranging from 0-600.
- **Higher scores** indicate a lesser degree of handicap or **greater degree of social and community participation.**

Ex. PHYSICAL INDEPENDENCE scoring with CHART

Revised Craig Handicap Assessment and Reporting Technique Scoring Form

		PHYSICAL INDEPENDENCE
1. How many hours in a typical 24-hour day do you have someone with you to provide physical assistance for personal care activities such as eating, bathing, dressing, toileting and mobility? _____ hours paid assistance _____ hours unpaid (family, others)	Total the hours of paid and unpaid care.	_____ +
2. Not including any regular care as reported above, how many hours in a <u>typical month</u> do you occasionally have assistance with such things as grocery shopping, laundry, housekeeping, or infrequent medical needs because of the disability? _____ hours per month	Divide the hours of occasional care by 30. Add the sums of "A" and "B".	(_____ / 30) = _____
3. Who takes responsibility for instructing and directing your attendants and/or caregivers? _____ Self _____ Someone Else _____ Not applicable, does not use attendant care	If the respondent instructs and directs his/her own attendants or caregivers, multiply the answer of "C" by 3. If someone other than the respondent instructs and directs the attendants or care givers, multiply the answer of "C" by 4.	X 3 or 4 = _____
	Subtract the total in "D" from 100.	100 minus _____ sum from "D" above = _____

PHYSICAL INDEPENDENCE SCORE

In the **Physical Independence** dimension, a score of 100 indicates no handicap in an individual's ability to sustain a customarily effective independent existence. The need for regular or periodic assistance for activities which used to be performed independently is indicative of some degree of handicap.

SCORE COMPUTATION:

- Add the hours of paid to unpaid care in Question #1.
- Divide the hours of occasional care in Question #2 by 30. This computation results in an equivalent of hours per day of occasional care.
- Add the sums of "A" and "B".
- If the respondent instructs and directs his/her own attendants or caregivers (Question #3), multiply the answer of "C" by 3. If someone other than the respondent instructs and directs the attendants or care givers, multiply the answer of "C" by 4.
- Subtract the total in "D" from 100. The remainder will be the **Physical Independence** dimension score.

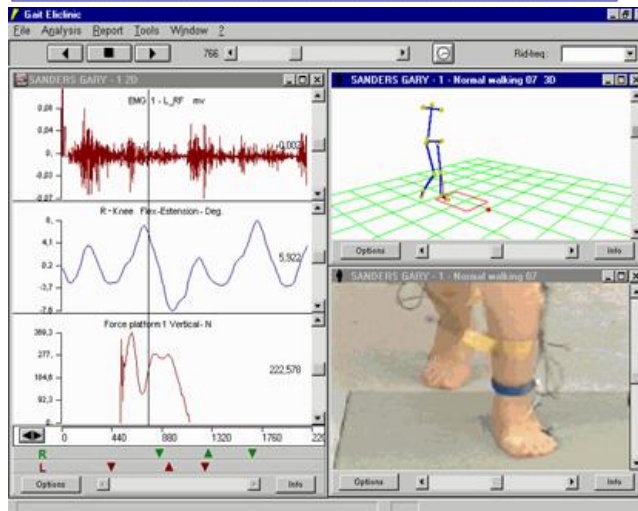
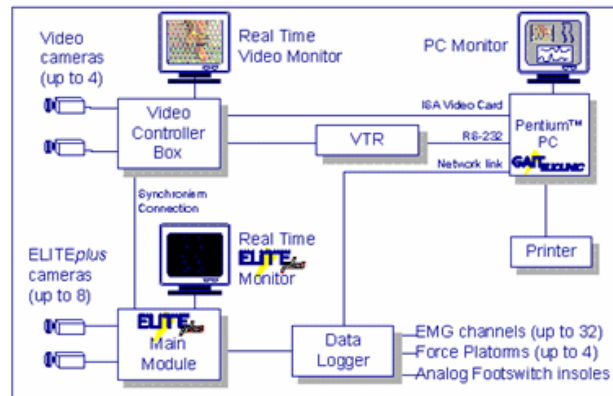
Assessment scales for the quality of life

- Assess the impact of disease from the patient's point of view and measure the parameters related to the perception that the patient has of him/herself, from the psychological and physical point of view
- Most common scales:
 - SIP (Sickness Impact Profile)
 - SF-36 (36-Item Short-Form)
 - MRFA (Medical Rehabilitation Follow Along)

Characteristics of assessment scales

- **Reliability:** how much the measure is free from errors
Repeatability: how much the measure can be reproduced, when executed more times or by more examiners, appropriately and in similar conditions
- **Inter-examiner reproducibility :** level of overlapping between scores obtained by different trained examiners
- **Test-retest reproducibility:** level of variability of scores obtained in repeated measurements in a so-called stable status

Motion analysis: quantitative study of human movement



- **Kinematic data:** human movements, measured with motion capture systems;
- **Dynamic data:** forces and angular moments, measured with force platforms and other sensors;
- **Electro-myographic (EMG) data:** electrical signals of muscle activations, measured with electrodes

Summary

- Learnt some terminology about human anatomy
- Anatomy and role of vertebral column
- Spinal cord injuries and related paralyses
- Scales for assessment of
 - Impairment
 - Disability
 - Handicap
 - Quality of life